

RESEARCH ARTICLE

Closed-form design formulas of TMDI for suppressing vortex-induced vibration of bridge structures

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Summary

Tuned mass damper inerter (TMDI) is a promising device for structural vibration control. Existing design formulas for TMDI are derived based on an undamped single-degree-of-freedom primary structure. When applied to vortex-induced vibration (VIV) control of bridges, it may lead to a suboptimal design of TMDI, since VIV is characterized by nonlinear aeroelastic effect and sensitive to the structural damping. This study proposes closed-form design formulas of TMDI that are suitable for VIV control of bridges. Governing equations of the bridge-TMDI system are first established in modal coordinate. Based on these equations, the equivalent damping of TMDI to the bridge is derived. Then, the design formulas for TMDI frequency and damping ratio with predetermined TMDI mass, inertance, and inerter arrangement are developed to achieve the required equivalent damping of TMDI. The reliability of the proposed formulas is validated by utilizing the wind tunnel experimental data of a bridge. It is found that the effect of TMDI to the bridge can be treated as an increase of the structural damping. It is also found that the TMDI parameters will betray the optimum values if the VIV force model fails to correctly predict the required equivalent damping of TMDI. To eliminate this disadvantage, some design suggestions are presented. The comparison with existing TMDI design formulas and discussions on the application scope of the proposed formulas are also conducted. The proposed formulas can achieve a better control efficiency for the same predetermined TMDI parameters and have high accuracy within the scope of practical engineering applications.

KEYWORDS

design formulas, long-span bridge, TMDI, vibration control, vortex-induced vibration

1 | INTRODUCTION

Vortex-induced vibration (VIV) is a commonly occurred fluid-structural interaction phenomenon for flexible bridges.¹ It suggests that all bridge types could suffer the threats of VIV. Due to the self-limiting property of amplitude, VIV hardly leads to the catastrophic collapse of the bridge as flutter does. However, a cyclic large-amplitude or high-frequency oscillation threatens the serviceability and fatigue life of the bridge and may cause the failure of structural elements. Besides, VIV of the bridge is prone to occur at moderate or low wind velocities which can be easily achieved at the site of the bridge. VIV suppression therefore has received intensive concerns in the bridge and wind engineering communities.

To suppress the VIV of bridges, normally three countermeasures can be adopted²: (1) the aerodynamic countermeasure, such as tailoring the geometry of the section and shifting the handrail and/or maintenance track, or adding aerodynamic attachment devices; (2) the mechanical countermeasure, such as tuned mass dampers (TMDs); and (3) the structural countermeasure, such as adding intermediate support to improve the structural stiffness. Although the aerodynamic countermeasure is very cheap and practically convenient, it is time consuming and labor intensive to conduct the wind tunnel tests before obtaining an effective countermeasure. Sometimes, these countermeasures even fail to mitigate the VIV in real bridges due to the gap of the Reynolds number in the tests and the real situation. The structural countermeasure is usually complicated and has limited improvement in reducing the VIV effects. Comparatively, TMD is considered to be one of the simplest and most effective devices since VIV is always characterized by the single-degree-of-freedom (SDOF) oscillation. It is especially applicable for bridges by installing the TMDs inside the box girder, which has insignificant influence on the serviceability as well as esthetics of the bridge. Thus, TMD is often employed as an alternative when performance of aerodynamic countermeasures is unsatisfactory or as a supplemental plan to aerodynamic countermeasures to further enhance the control efficiency.

TMD is achieved using a comparatively lightweight mass component mounted in the primary structure via the spring and dashpot elements. The fundamental idea behind TMD is to transfer vibration energy from the primary structure to the secondary system by optimally tuning the TMD parameters (mass, damping, and stiffness). In such a way, the secondary mass undergoes large vibrations and dissipates the energy absorbed from the primary structure by the damper to reduce the structure's response. To maximize the control efficiency of TMD, the optimum design of TMD parameters is of vital importance. The earliest works on TMD design were conducted by Den Hartog³ for a SDOF undamped primary structure subjected to harmonic excitations, and Crandall and Mark⁴ for random base excitations. Fujino and Abé⁵ then derived the optimal design formulas of TMD for lightly damped SDOF primary structures under various types of loading (free vibration, random excitation, harmonic excitation, and self-excitation). Bakre and Jangid⁶ numerically examined the optimum TMD parameters attached to a damped SDOF primary structure at various combinations of objective functions and excitations, and proposed explicit expressions of TMD parameters through fitting the numerical results. Ghosh and Basu⁷ derived a closed-form formula for the optimum frequency of TMD attached to a lightly damped SDOF primary structure by introducing the fixed-point theory. In addition to the above-mentioned design formulas, there also exist enormous studies that utilized different numerical solutions and optimization strategies to identify TMD's optimum parameters (e.g., Hadi and Arfiadi,⁸ Leung and Zhang,⁹ Bekdas and Nigdeli,¹⁰ and Jangid¹¹). It should be noted that, in most previous studies, damping for the structure was either ignored or only a light linear structural damping was considered.

It is known that VIV of a structure is very sensitive to a non-dimensional mass-damping parameter, that is, the Scruton number (defined as $S_c = 2m\xi/(\rho D^2)$, where m is structural mass per unit length, ξ is the structural damping ratio, ρ is the air density, and D is the structural reference size). The contribution of TMD to the bridge is equivalent to increasing the damping or changing the Scruton number of the bridge. Thus, to sufficiently suppress VIV of the bridge, the following two issues need to be clarified: (1) the required increment of Scruton number to achieve the design target (e.g., the required reduction percentage) and (2) the optimum TMD parameters (mass, damping and stiffness) to achieve this increment. In this process, the influence of aeroelastic effect of VIV should also be carefully considered, since a slight change in the aeroelastic effect can lead to a totally different VIV property of the structure, even though the Scruton number of the structure remains unchanged.

Xiang et al.¹² first derived the equivalent damping contribution of TMD to the bridge using the Krylov-Bogoliubov method, in which the aeroelastic effect of VIV was modeled by the Scanlan nonlinear empirical model.¹³ Larsen et al.¹⁴ compared the optimized TMD parameters obtained by considering and ignoring the influence of aeroelastic effect. The results indicated that the required minimum TMD mass to fully suppress the VIV of the bridge is approximately one third of that designed by ignoring the influence of aeroelastic effect. Dai et al.¹⁵ further investigated the optimized TMD parameters using Scanlan linear and nonlinear empirical models. It showed that a much smaller TMD mass is required using Scanlan nonlinear model than that obtained by the linear model to achieve the same control efficiency.

Although TMD is an effective device to reduce VIV and has been applied to several bridges, it is challenging to apply it in the streamlined flat box girder of long-span bridges. This is because the VIV frequency of the long-span bridge is relatively low. To control it, the static deformation of the TMD springs could be quite large. The static deformation of TMD springs can be assessed by $L \approx 0.25/f_b^2$, in which f_b is the natural frequency of the bridge.¹⁶ Alternatively, advanced inerter-based vibration control devices provide a novel idea to extend the application of TMD. The inerter is a two-terminal device which can transform the lateral motion of the object into rotation. It is therefore able to achieve a much larger inertial force than that obtained by the object's physical mass.¹⁷ The most commonly studied inerter-based

vibration control devices are tuned mass damper inerter (TMDI),¹⁸ the tuned inerter damper (TID),¹⁹ the inerter-based dynamic vibration absorbers (IDVAs),²⁰ the tuned tandem TMDI,²¹ and the tuned liquid column damper inerter.²² Application studies of these devices can be found in high-rise buildings,^{23,24} adjacent buildings,²⁵ base-isolated systems,^{26,27} long-span bridges,²⁸ wind turbines,^{29,30} and offshore platforms.^{31,32} Ma et al.³³ recently give a comprehensive review on the applications of inerter-based devices in civil engineering structures.

Among these inerter-based vibration control devices, TMDI is especially appealing for the VIV control of long-span bridges. It consists of a conventional TMD and an inerter which is in parallel with the TMD spring and dashpot elements. The contribution of inerter during vibration is equivalent to offer a virtual mass to the TMD mass block (due to the mass amplification effect of inerter). The stiffness of TMDI spring is tuned based on the summation of TMD physical mass and inerter's virtual mass, while the static deformation of TMDI spring is only influenced by the gravity of TMD physical mass. In this sense, TMDI allows a much shorter static deformation than that of conventional TMD using the same mass ratio and thus can be installed inside of the box girder more conveniently to control the low mode vertical VIV of long-span bridges. Xu et al.²⁸ first studied the control performance of TMDI for VIV of bridges. The results show that the static deformation of TMD springs can be significantly reduced through involving of inerter. Dai et al.³⁴ investigated the effect of inerter span length of TMDI, indicating that the control efficiency increases as inerter spans longer girder length. Furthermore, Xu et al.³⁵ compared different layouts of inerter-based dynamic vibration absorbers for bridge VIV control. Dai et al.³⁶ utilized a maxwell element to replace the liner dashpot in TMDI and examined its performance in bridge VIV control. Recently, Chen et al.³⁷ further compared the performance of different layouts of IDVAs for bridge VIV control.

Similar to the conventional TMD device, the use of TMDI also requires an optimum design of parameters before obtaining the best performance. Marian and Giaralis¹⁸ first derived the closed-form design formulas of TMDI for undamped SDOF primary structures subjected to stochastically support excitations by minimizing the displacement variance of the primary structure. Later, Hu and Chen²⁰ proposed closed-form design formulas of TMDI for undamped SDOF primary structures through minimizing the H_2 and H_∞ norms of the frequency response function of the system. Zhang and Fitzgerald³⁰ proposed closed-form expressions for the optimal frequency and damping ratios of blade-mounted TMDIs in wind turbines based on the fixed-point theory. The damping of the primary structure was ignored, and the external force was assumed to be harmonic. In addition, the TMDIs were also designed using the TMD design formulas by assuming that the mass of TMD equals to the sum of TMDI mass block and inerter's virtual mass. This method was employed to design the TMDI in VIV control of a high-rise building by Giaralis and Petrini²³ and in the VIV control of a short-span bridge by Dai et al.³⁶ It is noteworthy that the above-mentioned formulas neglect the structural damping and cannot consider the influence of aeroelastic effect during VIV. Xu et al.²⁸ designed the TMDI for bridge VIV control through numerical optimization. The influence of aeroelastic effect was introduced via a constrained minimization of the deck's response. However, the numerical optimization method is unable to give an explicit expression of TMDI parameters with respect to structural parameters, which is inconvenient for use by bridge designers and difficult to be incorporated into design standards.

This paper aims to develop the closed-form design formulas of TMDI for bridge VIV control. Firstly, the generalized governing equations of the bridge-TMDI system considering the installation position of TMDI and inerter span length are established in modal coordinate in section 2. Based on the governing equations, the equivalent damping effect of TMDI on the bridge is derived. In section 3, the design formulas of TMDI frequency and damping ratio with predetermined mass ratio, inertance ratio, and inerter arrangement are derived to achieve the required equivalent damping of TMDI. In section 4, the accuracy of the design formulas is validated through wind tunnel experimental data, and the influence of VIV model parameters on TMDI design is also investigated. In section 5, a comparison with existing design formulas of TMDI is conducted, and the application scope of the proposed formulas is also discussed.

2 | GOVERNING EQUATIONS OF BRIDGE-TMDI SYSTEM

A bridge equipped with a TMDI is shown in Figure 1. In this arrangement, a generalized TMDI configuration is considered, in which the two ends of the inerter (b in Figure 1) spans at different locations of the bridge deck (as shown, one end of the inerter is at section A, while the other end is at section B) to generate different accelerations between the two ends of the device, since the control force provided by an inerter is proportional to the relative accelerations between its two ends. In the figure, $y(x, t)$ is vertical displacement of the deck at spanwise position x . x_{tmdi} denotes the spanwise location where the TMDI mass block is installed. x_{inerter} is the spanwise location where the other terminal of

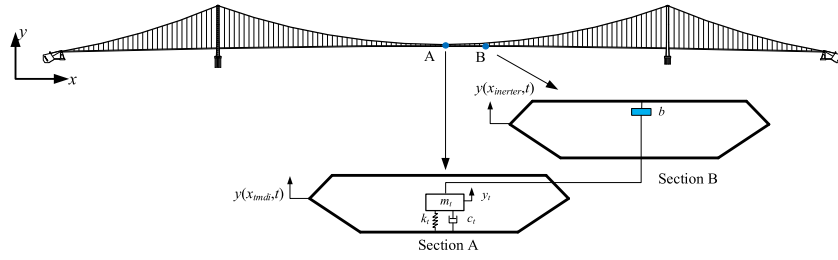


FIGURE 1 Bridge-TMDI system

the inerter is linked to ($x_{\text{tmdi}} = x_{\text{inserter}}$ if the inerter spans zero length). m_t , k_t , c_t , and b are the mass, stiffness, damping, and inertance of the TMDI, respectively. $y_t(t)$ is the displacement of TMDI mass block.

According to the field measurements,^{38–40} VIV often occurs at a single mode of the bridge. Thus, the displacement of the bridge can be expressed as $y(x, t) = \phi_j(x) \cdot q(t)$, where $\phi_j(x)$ and $q(t)$ are the modal shape function and generalized coordinate of the j th bridge mode, respectively. As to the motion of TMDI, it is normally assumed that the inclusion of TMD or TMDI to the bridge has little influence on its modal properties, since the mass of the TMD or TMDI is quite small compared to the bridge. Thus, the motion of TMDI, $y_t(t)$, can be treated as an additional DOF to the bridge. The equations of motion for the bridge and the TMDI can then be expressed as

$$M[\ddot{q}(t) + 2\omega_b \xi_b \dot{q}(t) + \omega_b^2 q(t)] = F(U, q, \dot{q}, t) + f_b \phi_j(x_{\text{inserter}}) + f_t \phi_j(x_{\text{tmdi}}) \quad (1.1)$$

$$m_t \ddot{y}_t(t) = -f_b - f_t \quad (1.2)$$

$$f_b = b [\ddot{y}_t(t) - \phi_j(x_{\text{inserter}}) \ddot{q}(t)] \quad (1.3)$$

$$f_t = k_t [y_t(t) - \phi_j(x_{\text{tmdi}}) q(t)] + c_t [\dot{y}_t(t) - \phi_j(x_{\text{tmdi}}) \dot{q}(t)] \quad (1.4)$$

where $M = \int_0^L m \phi_j^2(x) dx$ is modal mass of the bridge with m as the equivalent modal mass of the deck per unit length; L is the total length of the bridge; and ω_b and ξ_b are the modal frequency and damping ratio of the bridge, respectively. $F(U, q, \dot{q}, t)$ denotes the VIV force imposed on the bridge mode, which is often modeled as a function of wind velocity as well as the displacement, velocity, and/or acceleration of bridge deck.¹³

By introducing $\Delta q_t(t) = y_t(t) - \phi_j(x_{\text{tmdi}}) q(t)$ as the relative displacement between TMDI mass block and the deck, and $\Delta \phi = \phi_j(x_{\text{inserter}}) / \phi_j(x_{\text{tmdi}})$ as the modal shape ratio between the two terminals of the inerter, the above governing equations can be rewritten as

$$M[1 + \phi_j(x_{\text{tmdi}})^2 \mu + \phi_j(x_{\text{tmdi}})^2 (1 - \Delta \phi)^2 \beta] \ddot{q} + 2M\omega_b \xi_b \dot{q} + M\omega_b^2 q + M[\phi_j(x_{\text{tmdi}}) \mu + \phi_j(x_{\text{tmdi}}) (1 - \Delta \phi) \beta] \Delta \ddot{q}_t = F(U, q, \dot{q}, t) \quad (2.1)$$

$$[\phi_j(x_{\text{tmdi}}) \mu + \phi_j(x_{\text{tmdi}}) (1 - \Delta \phi) \beta] \ddot{q} + (\mu + \beta) \Delta \ddot{q}_t + 2(\mu + \beta) \omega_t \xi_t \Delta \dot{q}_t + (\mu + \beta) \omega_t^2 \Delta q_t = 0 \quad (2.2)$$

where $\mu = m_t / M$ is defined as the mass ratio of TMDI; $\beta = b / M$ is the inertance ratio; $\omega_t = \sqrt{k_t / (m_t + b)}$ is the vibration frequency of TMDI; and $\xi_t = c_t / [2(m_t + b) \omega_t]$ is the damping ratio of TMDI.

It is noticeable that the above equation is a generalized form for bridge-TMDI system, where the TMDI can be installed at any spanwise position along the deck, and the inerter can span to any different position from the TMDI mass block. If the TMDI is installed at the generalized coordinate of the bridge mode and the inerter spans zero length from the TMDI mass block, that is, $\phi_j(x_{\text{inserter}}) = \phi_j(x_{\text{tmdi}}) = 1$, the above function degrades to the one established in Xu et al.²⁸

2.1 | VIV response of the deck without TMDI

To calculate the VIV of the bridge, the Krylov-Bogoliubov method is normally adopted.¹³ It assumes that the vibration amplitude $A(t)$ and phase $\psi(t)$ of the bridge are slowly varying functions of time t , which change very little over one period of oscillation. Thus, the displacement of generalized coordinate and the corresponding velocity can be expressed as $q(t) = A(t)\cos[\omega t - \psi(t)]$ and $\dot{q}(t) = -A(t)\omega\sin[\omega t - \psi(t)]$, respectively.

Combining with Equation (2.1) and neglecting the influence of TMDI, the governing equations of the bridge under VIV can be expressed as

$$\dot{A}(t)\cos[\omega t - \psi(t)] + A(t)\dot{\psi}\sin[\omega t - \psi(t)] = 0 \quad (3.1)$$

$$\begin{aligned} F(U, q, \dot{q}, t)/M = & -\dot{A}(t)\omega\sin[\omega t - \psi(t)] - \omega^2 A(t)\cos[\omega t - \psi(t)] + \dot{\psi}A(t)\omega\cos[\omega t - \psi(t)] + \omega_b^2 A(t)\cos[\omega t - \psi(t)] \\ & - 2\omega_b\xi_b A(t)\omega\sin[\omega t - \psi(t)] \end{aligned} \quad (3.2)$$

Solving the equations above, it is easy to get

$$\dot{A}(t) = -\frac{1}{\omega} \left\{ \begin{aligned} & F(U, q, \dot{q}, t)/M + 2\omega_b\xi_b A(t)\omega\sin[\omega t - \psi(t)] \\ & + A(t)(\omega^2 - \omega_b^2)\cos[\omega t - \psi(t)] \end{aligned} \right\} \sin[\omega t - \psi(t)] \quad (4.1)$$

$$\dot{\psi}(t) = \frac{1}{\omega A(t)} \left\{ \begin{aligned} & F(U, q, \dot{q}, t)/M + 2\omega_b\xi_b A(t)\omega\sin[\omega t - \psi(t)] \\ & + A(t)(\omega^2 - \omega_b^2)\cos[\omega t - \psi(t)] \end{aligned} \right\} \cos[\omega t - \psi(t)] \quad (4.2)$$

Since $A(t)$ and $\psi(t)$ are slowly time varying, their values can be averaged in a single period of vibration. In addition, the vibration amplitude and phase are constants at steady-state of VIV, thus, the time derivatives of $A(t)$ and $\psi(t)$ equal to zero. The above equations can be rearranged as

$$\int_0^{2\pi} F(U, q, \dot{q}, t)\sin\tau d\tau = -2\pi A\omega M\omega_b\xi_b \quad (5.1)$$

$$\int_0^{2\pi} F(U, q, \dot{q}, t)\cos\tau d\tau / (\pi AM) = \omega_b^2 - \omega^2 \quad (5.2)$$

where $\tau = \omega t - \psi$ is defined as the phase of vibration.

Normally, the structural stiffness is higher than the aeroelastic stiffness by several orders of magnitude, and the latter is thus practically negligible.⁴¹ Then, the above function can be rewritten as

$$\int_0^{2\pi} F(U, q, \dot{q}, t)\sin\tau d\tau = -2\pi A\omega M\omega_b\xi_b \quad (6.1)$$

$$\int_0^{2\pi} F(U, q, \dot{q}, t)\cos\tau d\tau / (\pi AM) = \omega_b^2 - \omega^2 = 0 \quad (6.2)$$

As can be seen, since the aeroelastic stiffness is neglected, the vibration frequency of the bridge during VIV equals to its natural frequency.

2.2 | VIV response of the deck with TMDI

The VIV of the bridge with TMDI can also be calculated by the Krylov-Bogoliubov method. The motions of the bridge and TMDI can be expressed as

$$q(t) = \Lambda_1 e^{i\omega t} \quad (7.1)$$

$$\Delta q_t(t) = \Lambda_2 e^{i\omega t} \quad (7.2)$$

where ω is the vibration frequency of the bridge-TMDI system; Λ_1 and Λ_2 are complex constants, denoting the vibration amplitudes of the bridge and TMDI, respectively.

Substituting Equations (7.1) and (7.2) into Equation (2.2), it yields

$$\frac{\Lambda_2}{\Lambda_1} = g + hi = \frac{\gamma^2 \phi_j(x_{\text{tmdi}}) [\mu + (1 - \Delta\phi)\beta] [(\mu + \beta)\Omega^2 - (\mu + \beta)\gamma^2]}{[(\mu + \beta)\Omega^2 - (\mu + \beta)\gamma^2]^2 + [2(\mu + \beta)\xi_t \gamma \Omega]^2} - \frac{\gamma^2 \phi_j(x_{\text{tmdi}}) [\mu + (1 - \Delta\phi)\beta] [2(\mu + \beta)\xi_t \gamma \Omega]}{[(\mu + \beta)\Omega^2 - (\mu + \beta)\gamma^2]^2 + [2(\mu + \beta)\xi_t \gamma \Omega]^2} i \quad (8)$$

where $\Omega = \omega_t / \omega_b$ denotes the ratio between natural frequencies of TMDI and the bridge; $\gamma = \omega / \omega_b$ is the ratio between the vibration frequency of the bridge with TMDI to that without TMDI.

Substituting $\Delta q_t(t) = \Lambda_2 q(t) / \Lambda_1$ into Equation (2.1), it yields

$$\ddot{q} + \omega_b^2 q = F(U, q, \dot{q}, t) / M + \omega^2 \phi_j^2(x_{\text{tmdi}}) [\mu + (1 - \Delta\phi)^2 \beta] q + \phi_j(x_{\text{tmdi}}) [\mu + (1 - \Delta\phi)\beta] \omega^2 g q + \phi_j(x_{\text{tmdi}}) [\mu + (1 - \Delta\phi)\beta] \omega h \dot{q} - 2\omega_b \xi_b \dot{q} \quad (9)$$

Assuming $q(t) = A(t) \cos[\omega t - \psi(t)]$ and $\dot{q}(t) = -A(t) \omega \sin[\omega t - \psi(t)]$, and following the same procedure presented in the above section, it has

$$\int_0^{2\pi} F(U, q, \dot{q}, t) \sin \tau d\tau = -2\pi A \omega M \omega_b \left\{ \xi_b + \frac{\gamma^4 \phi_j^2(x_{\text{tmdi}}) [\mu + (1 - \Delta\phi)\beta]^2 [(\mu + \beta)\xi_t \Omega]}{[(\mu + \beta)\Omega^2 - (\mu + \beta)\gamma^2]^2 + [2(\mu + \beta)\xi_t \gamma \Omega]^2} \right\} \quad (10.1)$$

$$\int_0^{2\pi} F(U, q, \dot{q}, t) \cos \tau d\tau / (\pi A M \omega_b^2) = -(\gamma^2 - 1) - \phi_j^2(x_{\text{tmdi}}) [\mu + (1 - \Delta\phi)^2 \beta] \gamma^2 - \frac{\gamma^4 \phi_j^2(x_{\text{tmdi}}) [\mu + (1 - \Delta\phi)\beta]^2 [(\mu + \beta)\Omega^2 - (\mu + \beta)\gamma^2]}{[(\mu + \beta)\Omega^2 - (\mu + \beta)\gamma^2]^2 + [2(\mu + \beta)\xi_t \gamma \Omega]^2} \quad (10.2)$$

Again, if the aeroelastic stiffness is neglected, the above function can be rewritten as

$$\int_0^{2\pi} F(U, q, \dot{q}, t) \sin \tau d\tau = -2\pi A \omega M \omega_b \left\{ \xi_b + \frac{\gamma^4 \phi_j^2(x_{\text{tmdi}}) [\mu + (1 - \Delta\phi)\beta]^2 [(\mu + \beta)\xi_t \Omega]}{[(\mu + \beta)\Omega^2 - (\mu + \beta)\gamma^2]^2 + [2(\mu + \beta)\xi_t \gamma \Omega]^2} \right\} \quad (11.1)$$

$$\frac{\gamma^4 \phi_j^2(x_{\text{tmdi}}) [\mu + (1 - \Delta\phi)\beta]^2 [(\mu + \beta)\Omega^2 - (\mu + \beta)\gamma^2]}{[(\mu + \beta)\Omega^2 - (\mu + \beta)\gamma^2]^2 + [2(\mu + \beta)\xi_t \gamma \Omega]^2} + (\gamma^2 - 1) + \phi_j^2(x_{\text{tmdi}}) [\mu + (1 - \Delta\phi)^2 \beta] \gamma^2 = 0 \quad (11.2)$$

Similar to Equation (6), Equation (11) determines the steady-state VIV amplitude of the bridge (with TMDI) and the vibration frequency of the bridge-TMDI system. Comparing Equation (11.1) with Equation (6.1), it can be noted that the contribution of TMDI is equivalent to increasing the damping ratio of the bridge:

$$\xi_{\text{eq}} = \frac{\gamma^4 \phi_j^2(x_{\text{tmdi}}) [\mu + (1 - \Delta\phi)\beta]^2 [(\mu + \beta)\xi_t \Omega]}{[(\mu + \beta)\Omega^2 - (\mu + \beta)\gamma^2]^2 + [2(\mu + \beta)\xi_t \gamma \Omega]^2} \quad (12)$$

3 | DESIGN FORMULAS OF TMDI PARAMETERS

If an admissible bridge VIV amplitude is determined by following the suggestions specified in the wind resistance codes, then the required increment of damping by TMDI can be calculated by using VIV force models with model parameters captured by wind tunnel experiment. According to Equation (12), the equivalent damping of TMDI is determined by the following parameters: TMDI mass ratio μ , TMDI inertance ratio β , TMDI frequency ω_t , TMDI damping ratio ξ_t , TMDI installation position $\phi_j(x_{\text{tmdi}})$, and the inerter span parameter $\Delta\phi$. Normally, the values of mass ratio, inertance ratio, installation position, and inerter span are predetermined. Thus, the parameters to be designed are ω_t and ξ_t .

If the required equivalent damping value ξ_{eq} is given, the TMDI parameters are governed by the following equations:

$$\xi_{\text{eq}} = \frac{\gamma^4 \phi_j^2(x_{\text{tmdi}}) [\mu + (1 - \Delta\phi)\beta]^2 [(\mu + \beta)\xi_t \Omega]}{[(\mu + \beta)\Omega^2 - (\mu + \beta)\gamma^2]^2 + [2(\mu + \beta)\xi_t \gamma \Omega]^2} \quad (13.1)$$

$$\frac{\gamma^4 \phi_j^2(x_{\text{tmdi}}) [\mu + (1 - \Delta\phi)\beta]^2 [(\mu + \beta)\Omega^2 - (\mu + \beta)\gamma^2]}{[(\mu + \beta)\Omega^2 - (\mu + \beta)\gamma^2]^2 + [2(\mu + \beta)\xi_t \gamma \Omega]^2} + \phi_j^2(x_{\text{tmdi}}) [\mu + (1 - \Delta\phi)^2 \beta] \gamma^2 + (\gamma^2 - 1) = 0 \quad (13.2)$$

In the above equations, there are three unknown parameters to be solved: the frequency ratio of TMDI ($\Omega = \omega_t / \omega_b$), the damping ratio of TMDI ξ_t , and the frequency ratio of the bridge with and without TMDI ($\gamma = \omega / \omega_b$). Normally, the TMDI mass and inertance are quite small compared to the mass of the bridge. In this sense, the influence of TMDI on the natural frequency of the bridge is insignificant. Thus, it is assumed that the vibration frequency of the bridge remains unchanged with and without TMDI ($\gamma = 1$). Then, Equation (13) can be further expressed as

$$\tilde{S}_c = \frac{2m}{\rho D^2} \frac{\phi_j^2(x_{\text{tmdi}}) [\mu + (1 - \Delta\phi)\beta]^2 [(\mu + \beta)\xi_t \Omega]}{[(\mu + \beta)\Omega^2 - (\mu + \beta)]^2 + [2(\mu + \beta)\xi_t \Omega]^2} \quad (14.1)$$

$$\frac{\phi_j^2(x_{\text{tmdi}}) [\mu + (1 - \Delta\phi)\beta]^2 [(\mu + \beta)\Omega^2 - (\mu + \beta)]}{[(\mu + \beta)\Omega^2 - (\mu + \beta)]^2 + [2(\mu + \beta)\xi_t \Omega]^2} = -\phi_j^2(x_{\text{tmdi}}) [\mu + (1 - \Delta\phi)^2 \beta] \quad (14.2)$$

where $\tilde{S}_c = 2m\xi_{\text{eq}} / (\rho D^2)$ is defined as the required increment of Scruton number by TMDI.

Dividing Equation (14.1) by Equation (14.2), it has

$$\xi_t = -\frac{\rho D^2 \tilde{S}_c (\Omega^2 - 1)}{\phi_j^2(x_{\text{tmdi}}) [\mu + (1 - \Delta\phi)^2 \beta] 2m\Omega} \quad (15)$$

Submitting Equation (15) into Equation (14.2), it has

$$c_1 c_3^2 \left[\frac{(\tilde{S}_c \rho D^2)^2}{(m^2 c_1^2)} + 1 \right] \cdot \Omega^4 + \left\{ -2c_1 c_3^2 \left[\frac{(\tilde{S}_c \rho D^2)^2}{(m^2 c_1^2)} + 1 \right] + c_2 c_3 \right\} \cdot \Omega^2 + c_1 c_3^2 \left[\frac{(\tilde{S}_c \rho D^2)^2}{(m^2 c_1^2)} + 1 \right] - c_2 c_3 = 0 \quad (16.1)$$

$$(\Omega^2 - 1)^2 \left[1 + (\tilde{S}_c \rho D^2)^2 / (m^2 c_1^2) \right] \neq 0 \quad (16.2)$$

where the constant coefficients c_{1-3} can be expressed as

$$c_1 = \phi_j^2(x_{\text{tmdi}}) [\mu + (1 - \Delta\phi)^2 \beta] \quad (17.1)$$

$$c_2 = \phi_j^2(x_{\text{tmdi}})[\mu + (1 - \Delta\phi)\beta]^2 \quad (17.2)$$

$$c_3 = (\mu + \beta) \quad (17.3)$$

Equation (16.1) can be rewritten as

$$(\Omega^2 - 1) \left\{ c_1 c_3^2 \left[\frac{(\tilde{S}_c \rho D^2)^2}{(m^2 c_1^2)} + 1 \right] \cdot \Omega^2 - c_1 c_3^2 \left[\frac{(\tilde{S}_c \rho D^2)^2}{(m^2 c_1^2)} + 1 \right] + c_2 c_3 \right\} = 0 \quad (18)$$

Since $\Omega^2 = 1$ cannot meet the constraint condition in Equation (16.2), the following equation must be satisfied:

$$c_1 c_3^2 \left[\frac{(\tilde{S}_c \rho D^2)^2}{(m^2 c_1^2)} + 1 \right] \cdot \Omega^2 - c_1 c_3^2 \left[\frac{(\tilde{S}_c \rho D^2)^2}{(m^2 c_1^2)} + 1 \right] + c_2 c_3 = 0 \quad (19)$$

Neglecting the negative frequency, it has

$$\Omega = \sqrt{\frac{c_1 c_3^2 \left[\frac{(\tilde{S}_c \rho D^2)^2}{(m^2 c_1^2)} + 1 \right] - c_2 c_3}{c_1 c_3^2 \left[\frac{(\tilde{S}_c \rho D^2)^2}{(m^2 c_1^2)} + 1 \right]}} \quad (20)$$

and

$$\xi_t = \frac{c_2 c_3 \tilde{S}_c \rho D^2}{2c_1 m \sqrt{c_1 c_3^2 \left[\frac{(\tilde{S}_c \rho D^2)^2}{(m^2 c_1^2)} + 1 \right] - c_2 c_3} \cdot \sqrt{c_1 c_3^2 \left[\frac{(\tilde{S}_c \rho D^2)^2}{(m^2 c_1^2)} + 1 \right]}} \quad (21)$$

Equations (20) and (21) are the design formulas of TMDI for a given value of equivalent Scruton number of TMDI $\tilde{S}_c$. Thus, if a required increment of Scruton number by TMDI is determined, the values of TMDI frequency ω_t and TMDI damping ratio ξ_t can be obtained by using Equations (20) and (21). It should be noted that $\gamma = 1$ is assumed in the above process. That is to say, $\gamma = 1$ should be the solution of Equation (13.2) with parameters designed by Equations (20) and (21). However, since Equation (13.2) is a high-order algebraic equation, for a given set values of ω_t and ξ_t , there may exist other solutions of γ .

Equation (13.2) can be written as

$$a_1 \gamma^6 + a_2 \gamma^4 + a_3 \gamma^2 + a_4 = 0 \quad (22)$$

where

$$a_1 = c_3^2 + c_1 c_3^2 - c_3 c_2 \quad (23.1)$$

$$a_2 = -2c_3^2 \Omega^2 - c_3^2 + 4c_3^2 \Omega^2 \xi_t^2 + c_2 c_3 \Omega^2 - c_1 (2c_3^2 \Omega^2 - 4c_3^2 \Omega^2 \xi_t^2) \quad (23.2)$$

$$a_3 = 2c_3^2\Omega^2 + c_1c_3^2\Omega^4 + c_3^2\Omega^4 - 4c_3^2\Omega^2\xi_t^2 \quad (23.3)$$

$$a_4 = -c_3^2\Omega^4 \quad (23.4)$$

By introducing $R = \gamma^2$ and $R = \tilde{R} - a_2/(3a_1)$, Equation (22) can be rewritten as

$$\tilde{R}^3 + p\tilde{R} + q = 0 \quad (24)$$

where

$$p = \frac{3a_1a_3 - a_2^2}{3a_1^2} \quad (25.1)$$

$$q = \frac{27a_1^2a_4 - 9a_1a_2a_3 + 2a_2^3}{27a_1^3} \quad (25.2)$$

The above cubic equation can be solved by using the Cardano's formula. The property of the roots is determined by $\Delta = (q/2)^2 + (p/3)^3$. When $\Delta < 0$, Equation (24) has a real root and a pair of conjugate imaginary roots; when $\Delta = 0$, Equation (24) has three real roots (including one double root); and when $\Delta > 0$, Equation (24) has three unequal real roots.

If $R = \gamma^2$ has more than one real root, the TMDI has more than one natural frequency. However, since the TMDI is a linear system, it can only have one natural frequency in real practice. Thus, a constraint condition should be added to Equations (20) and (21), that is, $\Delta = (q/2)^2 + (p/3)^3$ must be less than zero. If this constraint condition does not satisfy, it indicates that the predetermined values of mass ratio μ , inertance ratio β , TMDI installation position $\phi_j(x_{\text{tmdi}})$, and inerter span $\Delta\phi$ cannot achieve the required value of $\tilde{S}_c$ through optimum tuning of ω_t and ξ_t . In this sense, the predetermined parameters should be reset or the required increment of Scruton number $\tilde{S}_c$ should be reduced. To better illustrate the use of these equations in TMDI design, a design flowchart is shown in Figure 2.

4 | VALIDATION OF THE DESIGN FORMULAS

4.1 | Validation through comparison with TMD design formulas

TMDI is often treated as a generalization of the conventional TMD. It degrades to TMD once the inertance becomes zero. If the inertance of TMDI is assumed to be null, the governing equations of TMDI for bridge VIV control (Equation 13) become

$$\xi_{\text{eq}} = \frac{\gamma^4[\mu\xi_t\Omega]}{[\Omega^2 - \gamma^2]^2 + [2\xi_t\gamma\Omega]^2} \quad (26.1)$$

$$\frac{\gamma^4[\mu\Omega^2 - \mu\gamma^2]}{[\Omega^2 - \gamma^2]^2 + [2\xi_t\gamma\Omega]^2} + \mu\gamma^2 + (\gamma^2 - 1) = 0 \quad (26.2)$$

Dai et al.¹⁵ optimized the TMD parameters (Ω and ξ_t) using a numerical optimization algorithm. For a given set of TMD parameters, the frequency ratio of the bridge (γ) is firstly calculated through solving Equation (26.2). Then the equivalent damping of TMD (ξ_{eq}) is calculated by Equation (26.1). The optimum values of TMD parameters can generate the maximum value of ξ_{eq} . Since Equation (26.2) is a high-order algebraic equation with respect to γ , the positive root that produces the smallest equivalent damping ratio is selected as the true value of γ . This is because the high equivalent damping ratio is normally ambiguous and unstable.

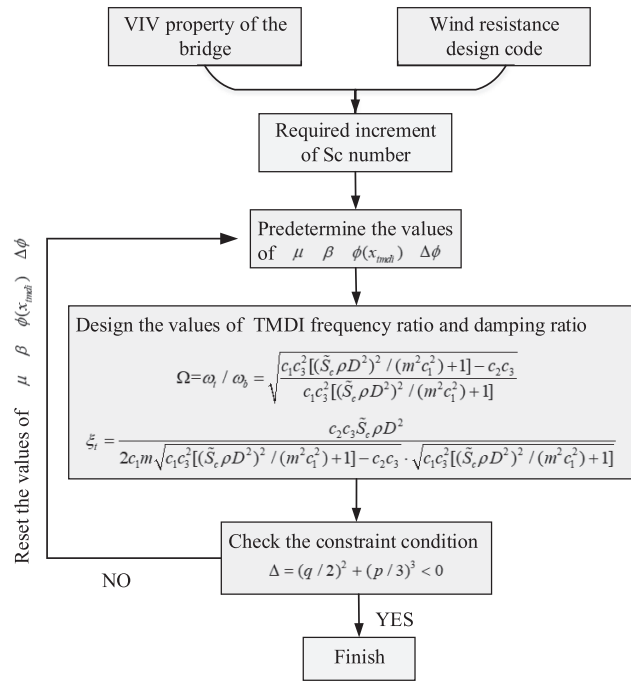


FIGURE 2 Design flowchart of TMDI for bridge VIV suppression using the proposed formulas

The calculated TMD parameters using the proposed formulas in this study (by setting the inertance related parameters in Equations (20) and (21) to zero) are compared with the results in Dai et al.,¹⁵ and the results are shown in Figure 3. As can be seen, the calculated curves of frequency ratio, damping ratio, and equivalent damping ratio for TMD using the proposed closed-form design formulas almost coincide with those obtained using numerical optimization, suggesting that the proposed formulas have enough accuracy and the hypothesis for deriving these equations is reasonable. However, it is worth noting that, Dai et al.¹⁵ only proposed the design formulas for the TMD system but not the TMDI system as investigated in the present study.

4.2 | Validation through wind tunnel experimental data

In this section, wind tunnel experimental data are adopted to further validate the efficiency and accuracy of the developed TMDI design formulas. The experiment was conducted in TJ-1 wind tunnel at Tongji University China, and the results were reported in a former study by the first author.¹ The VIV properties of the cross-sectional model at several different Scruton numbers were measured, and they are shown in Figure 4.

In the former study, the first author compared the capabilities of some commonly used VIV empirical models. It indicates that the Larsen's generalized empirical model can better reproduce the nonlinear property of the VIV system compared to the other VIV models.¹ Thus, this model is adopted to simulate the VIV force $F(U, q, \dot{q}, t)$ in Equation (2), which can be expressed as⁴²:

$$F(U, q, \dot{q}, t) = \frac{1}{2} \rho U^2 (2D) \left[Y_1 \left(\alpha_2 - \alpha_v \varepsilon \left| \frac{q(t)}{D} \right|^{2\nu} \right) \frac{\dot{q}(t)}{U} \right] \quad (27)$$

where ρ is air density; U is wind velocity; D is deck height; $\alpha_2 = \int_0^L \phi_j(x)^2 dx$ and $\alpha_v = \int_0^L \phi_j(x)^2 \cdot |\phi_j(x)|^{2\nu} dx$ are the modal shape coefficients that reflect the three-dimensional effect of the bridge; Y_1 and ε are the linear and nonlinear aeroelastic damping parameters, respectively; and ν is the second nonlinear aeroelastic damping parameter.

The VIV model parameters were identified by using the decay-to-resonance displacement of the deck. The experimental data at $S_c = 24.5$ and $U/(\omega_b D) = 3.16$ are adopted for this case study. The identified parameters are $Y_1 = 19587$,

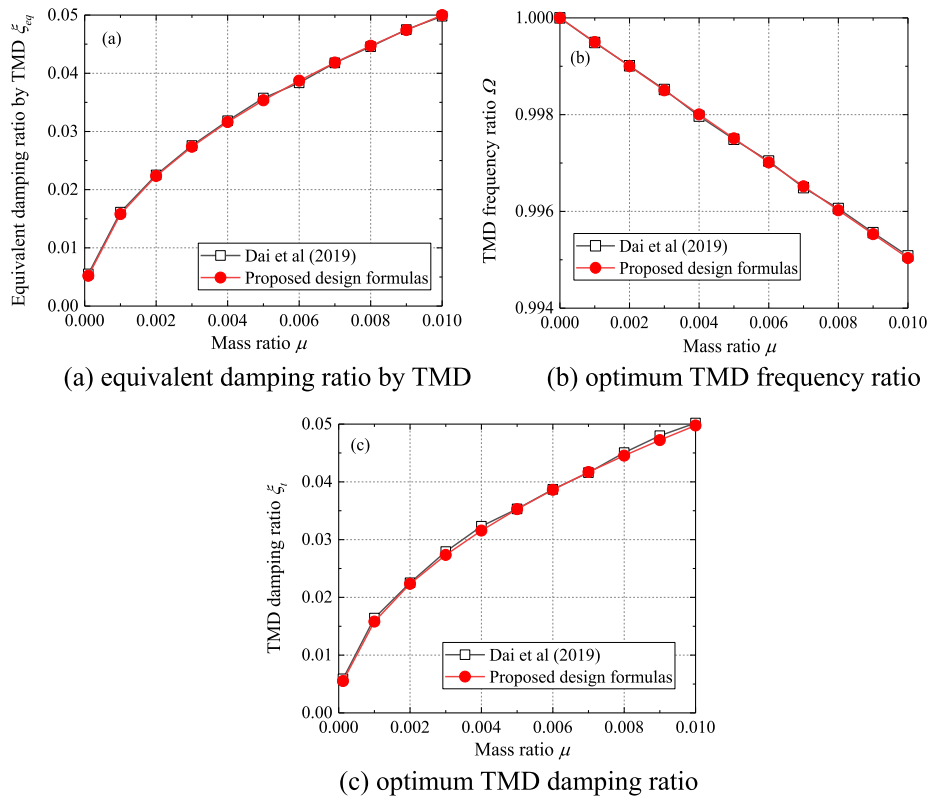


FIGURE 3 Comparison with numerical optimization method for TMD design

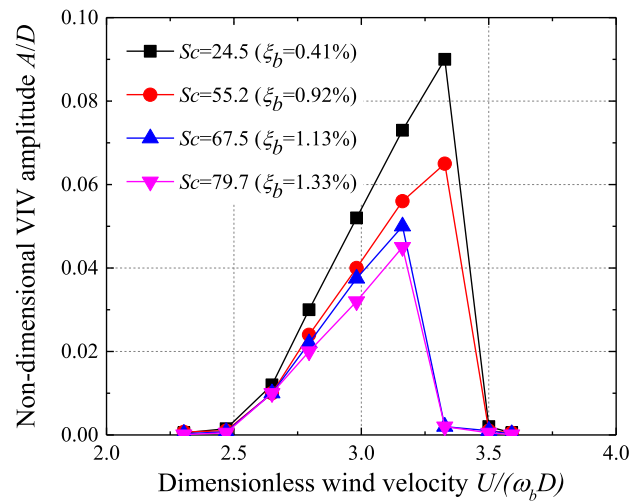


FIGURE 4 VIV properties of the sectional model at different Scruton numbers

$\varepsilon = 1.00525$, and $\nu = 0.000738$.¹ The simulated VIV response of the deck is compared with experimental result, as shown in Figure 5. As can be seen, the Larsen model can well reproduce the experimental result.

To ascertain the damping enhancement of TMDI, the VIV responses of the bridge with and without TMDI are predicted. The damping of the bridge system (i.e., the bridge without TMDI) as well as the total damping of the bridge-TMDI system are simultaneously increased to an identical value. Then, the VIV responses of the deck are calculated for comparison purpose. In particular, two damping values (i.e., 0.815% and 3.26%) are employed to achieve two Scruton numbers, and the corresponding results are shown in Figure 6.

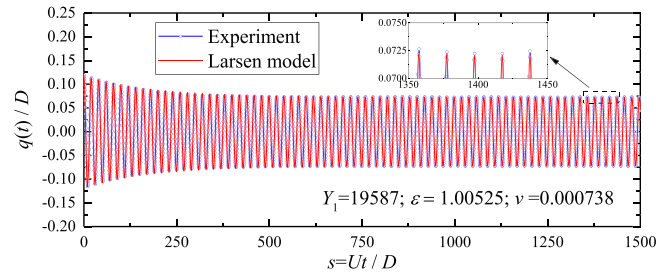


FIGURE 5 Simulated decay-to-resonance displacement of the deck

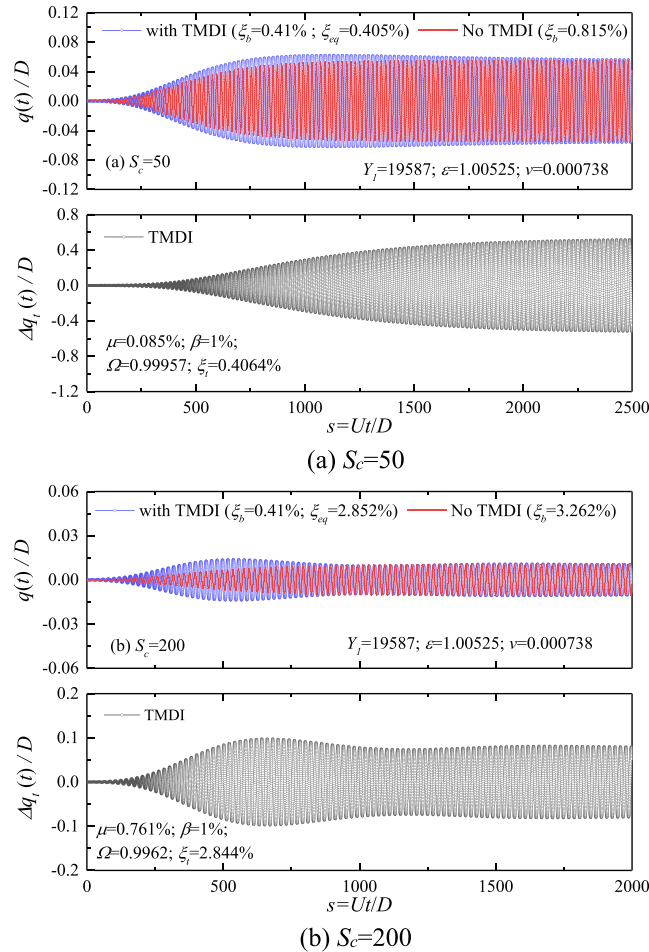


FIGURE 6 Validation of the damping enhancement effect of TMDI

Figure 6(a) shows the calculated response of the bridge at $S_c = 50$. In the calculation, the bridge deck is disturbed to a small amplitude position from the equilibrium position. After that, the deck response gradually increases due to the self-excitation property of VIV. With the increment of deck response, the TMDI is excited and the energy of the bridge system absorbed from the flow field begins to transfer to the TMDI system. With the increment of TMDI response, the deck response gradually approaches to a steady-state, where the energy absorbed from the flow field by the deck equals to those dissipated by the structural damping and the TMDI system. The steady-state response of the deck through directly increasing the structural damping is consistent with that of the bridge-TMDI system, indicating that the TMDI provides expected damping ratio to the bridge and its contribution to the bridge can be assessed using the equivalent damping described in Equation (12). Figure 6(b) shows the result at $S_c = 200$. With the increment of Scruton number, the VIV response of the bridge decreases. The steady-state response of the deck through directly increasing the

structural damping also matches well with the bridge-TMDI system, which can further validate the damping enhancement effect of TMDI to the bridge

It is noticeable that the simulated time histories in the transient part of grow-to-resonance by using TMDI and by directly increasing the structural damping are not the same. This is because the TMDI is static at the initial stage and is excited by the motion of the bridge. There is a phase lag between the bridge motion and the TMDI response, causing the TMDI needs some time to be adapted to the motion of the bridge. In this process, the damping effect of TMDI to the bridge is changing. Comparatively, when the structural damping is directly increased, the damping of the bridge system is a constant. However, the steady-state VIV amplitude is the concern of users and designers in bridge VIV assessment. The contribution of TMDI to the bridge can thus be treated as directly increasing the damping (or Scruton number) of the bridge.

4.3 | Influence of VIV model parameters

The equivalent damping of TMDI to the bridge derived in section 2.2 is independent of the VIV force model. However, it should be noted that if the required increment of Scruton number (i.e., the required TMDI equivalent damping) for suppressing the VIV of the bridge to an admissible value is calculated by the VIV force model, its influence should be paid attention to. This is because the VIV model parameters are sensitive to the Scruton number. For a given Scruton number at which the VIV model parameters are identified, it can well reproduce the experimental results. However, once the Scruton number varies from the one at which the model parameters are identified, the predicted response of the bridge will vary from the actual condition.¹

To investigate the influence of VIV model parameters on the design of TMDI, Equation (27) is substituted into Equation (11.1). Then, it has the steady-state VIV amplitude of the bridge-TMDI system versus Scruton number:

$$\frac{A}{D} = \frac{\sqrt{\frac{\pi\alpha_2 \left\{ 1 - \frac{2m\omega_b(\xi_b + \xi_{eq})}{\rho U D Y_1} \right\}}{\varepsilon\alpha_v \int_0^{2\pi} |\cos(\tau)|^{2v} \sin^2(\tau) d\tau}}}{\sqrt{\frac{\pi\alpha_2 \left\{ 1 - \frac{D\omega_b(S_c + \tilde{S}_c)}{U Y_1} \right\}}{\varepsilon\alpha_v \int_0^{2\pi} |\cos(\tau)|^{2v} \sin^2(\tau) d\tau}}} = \quad (28)$$

As can be seen, the VIV amplitude of the bridge-TMDI system is determined by not only the Scruton number but also the VIV model parameters.

The experimental data of the deck in section 4.1 at different Scruton numbers are utilized to identify the VIV model parameters. The selected Scruton numbers are $S_c = 24.5$ and $S_c = 55.2$ at $U/(\omega_b D) = 3.16$. The identified VIV model parameters are then adopted to calculate the VIV amplitude curves of the bridge-TMDI system versus Scruton number (by using Equation (28)). The results are shown in Figure 7. The solid and dashed lines denote the calculated results with VIV model parameters identified at $S_c = 24.5$ and $S_c = 55.2$, respectively. As can be seen, the model parameters identified at the above two Scruton numbers are different, which causes the difference between the two amplitude curves. The calculated VIV amplitude is identical with the experimental data under the Scruton number at which the model parameters are identified. However, at the other Scruton numbers, the calculated results vary from the experimental data. In addition, with the increment of Scruton number, the amplitude curves diverge more and more.

If an admissible VIV amplitude of $A/D = 0.045$ is employed, there exist two values of required increment of Scruton number from $S_c = 24.5$ using the predicted curves in Figure 7 (as shown by the arrows). To achieve these values, the required minimum values of TMDI mass ratio and inertance ratio are shown in Figure 8. The solid lines denote the required minimum combinations of TMDI mass and inertance ratios. The dashed lines denote the static deformation of TMDI springs (calculated by $L \approx 0.25/[f_b^2(1 + \beta/\mu)]$).

As can be seen, the required minimum mass and inertance ratios predicted by different VIV model parameters are not consistent. The VIV model parameters at $S_c = 24.5$ indicate that a larger TMDI mass ratio is needed; meanwhile, the static deformation of TMDI springs is also larger. The static deformation of TMDI springs can be reduced through increasing the inertance ratio, but the required mass ratio should be increased correspondingly. In this sense, the TMDI designed by this set of VIV model parameters needs a larger mass than that designed by VIV model parameters at $S_c = 55.2$. However, it should be noted that none of the above two curves predicted the actual value of required increment of Scruton number, as shown by the vertical dashed lines in Figure 7.

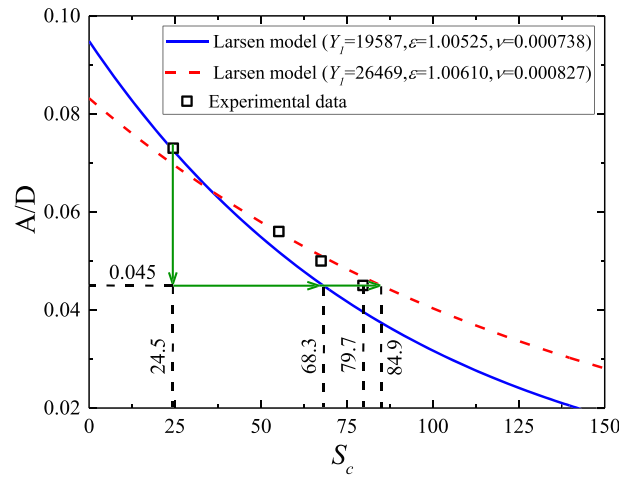


FIGURE 7 Predicted VIV amplitudes of the deck versus Scruton number

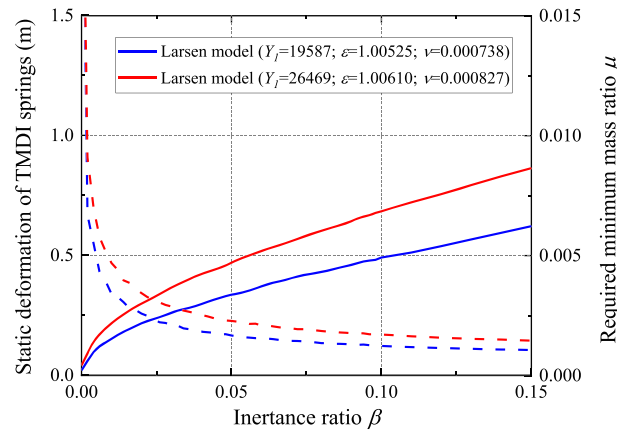


FIGURE 8 Influence of VIV model parameters on TMDI design

The above results indicate that the TMDI parameters depend on the accuracy of required equivalent Scruton number of TMDI. If the required equivalent Scruton number of TMDI is wrongly estimated, significant design errors will occur. To avoid this, it is recommended to conduct wind tunnel tests at various Scruton numbers and obtain the VIV amplitude curve directly through wind tunnel experiment. Then, the required equivalent Scruton number of TMDI can be directly obtained based on this experimental curve. The other way to reduce the design errors is to identify the VIV model parameters using experiment data at several different Scruton numbers. Then the VIV model parameters are fitted as a function of Scruton number. These fitted functions are then embedded into Equation (28) to obtain the amplitude curve and to calculate the required equivalent Scruton number of TMDI.

5 | FURTHER DISCUSSIONS

5.1 | Comparison with other design formulas of TMDI

To further validate the capacity of the proposed formulas, the TMDI design results are compared with those designed by the formulas in Marian and Giaralis¹⁸ and Hu and Chen.²⁰ In the derivation of the above formulas, the TMDI was installed at the maximum response position of the structure, and the inerter's span length was null, that is, $\phi(x_{\text{tmdi}}) = 1$ and $\Delta\phi = 1$. The above formulas are listed in Table I.

Figure 9(a) shows the equivalent damping ratio of TMDI at different values of mass and inertance ratios. For each set of mass and inertance ratios, the TMDI frequency ratio and damping ratio are calculated to maximize the equivalent

TABLE I Design formulas of TMDI parameters

	TMDI frequency ratio Ω	TMDI damping ratio ξ_t
Marian-Giavalis	$\frac{\sqrt{[\beta(\mu-1)+(2-\mu)(1+\mu)]}}{(1+\beta)\sqrt{2(1+\mu)}}$	$\frac{\sqrt{\beta+\mu}\sqrt{\beta(3-\mu)+(4-\mu)(1+\mu)}}{2\sqrt{2(1+\beta+\mu)[\beta(1-\mu)+(2-\mu)(1+\mu)]}}$
Hu-Chen (H_∞)	$\frac{\sqrt{\mu+(1+\mu)\beta}}{(1+\mu)\sqrt{\mu+\beta}}$	$\sqrt{\frac{3\mu^2}{8(1+\mu)(\mu+\beta)}}$
Hu-Chen (H_2)	$\sqrt{\frac{2\beta(1+\mu)+\mu(\mu+2)}{2(1+\mu)^2(\mu+\beta)}}$	$\sqrt{\frac{(\mu+\beta)+\Omega^2(\mu+\beta)(1+\mu)^2-\Omega^2\mu(2+\mu)-2\beta\Omega^2(1+\mu)}{4\Omega^2(\mu+\beta)(1+\mu)}}$

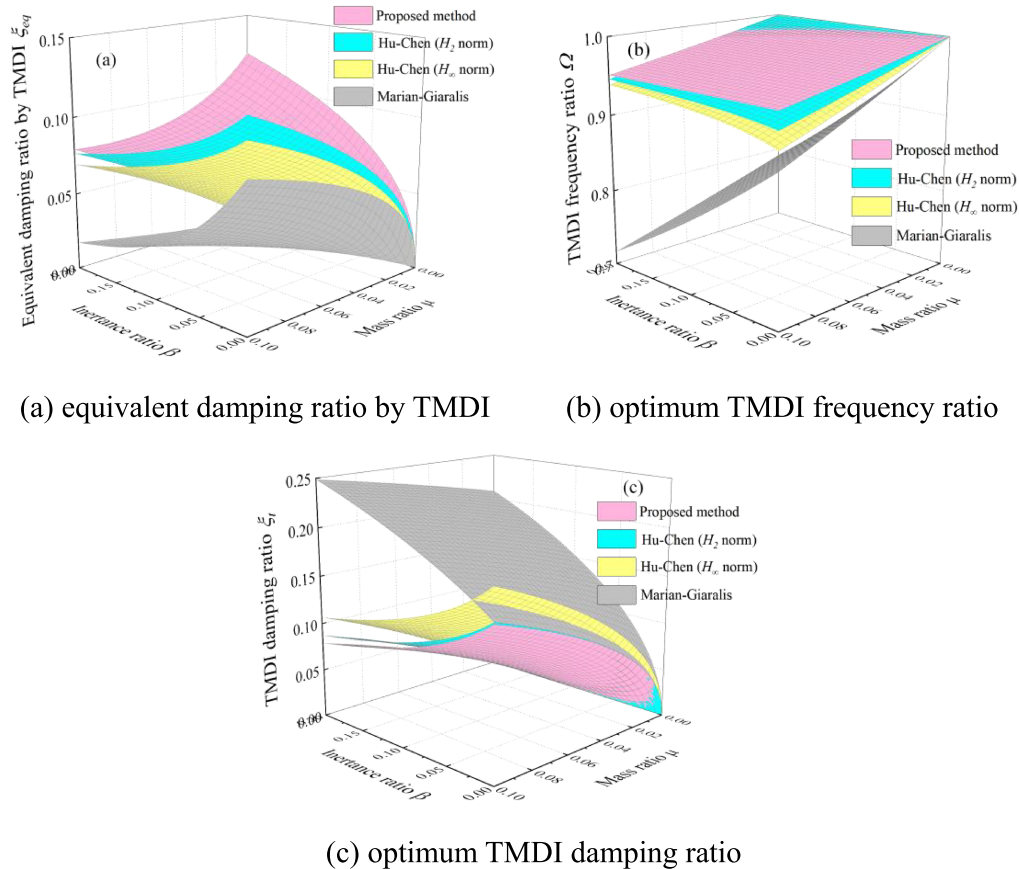


FIGURE 9 Comparison with other TMDI design formulas

damping of TMDI. As can be seen, the equivalent damping ratio of TMDI increases with the increment of mass ratio and decreases with the increment of inertance ratio, indicating that the control efficiency of TMDI increases with the increment of mass ratio and decreases with the increment of inertance ratio. This trend is consistent with the conclusions in previous studies.²⁸ For a given set of mass ratio and inertance ratio, the formulas in this study can obtain the largest values of equivalent damping ratio while the Marian-Giavalis formulas obtain the smallest values. The equivalent damping ratios obtained by Hu-Chen formulas are in between; and the results by using H_2 norm optimization are slightly better than those by using H_∞ norm optimization.

Figure 9(b) and 9(c) shows the optimized frequency ratios and damping ratios of TMDI. The formulas in this study and those of Hu-Chen have similar trends versus mass and inertance ratios, whereas the designed results by Marian-Giavalis formulas have a different trend versus inertance ratio. This difference may be due to the different arrangement of TMDIs. In Marian and Giavalis,¹⁸ the inerter is connected to the ground, whereas in this study and in Hu and Chen,²⁰ the inerter is connected in between the TMDI mass and the primary structure

The robustness of the bridge-TMDI system designed by different formulas is also compared through a sensitivity analysis against variations in structural parameters. Figure 10 shows the responses of the bridge ($\mu = 1\%; \beta = 1\%$). As can be seen, the control efficiency of the TMDI is sensitive to the variation of structural frequency and is less sensitive to the variation of structural damping. The TMDI designed by Marian-Giaralis formula has the worst control efficiency. In addition, the minimum VIV response of the bridge occurs at a structural frequency that is smaller than the one at which the TMDI is designed. This further validates that the Marian-Giaralis formula gets the suboptimum values of TMDI parameters. The bridge-TMDI systems designed by Hu-Chen formulas have similar bandwidth properties with the one designed by the proposed formula. Besides, the TMDI designed by H_2 norm can suppress the deck's response to the same magnitude as the one designed by the proposed formula. However, if we further compare the VIV responses of the bridge, the best control efficiency of TMDI designed by H_2 norm occurs at a slight smaller frequency value rather than the one at which the TMDI is designed, indicating that the H_2 norm does not generate the optimum TMDI parameters at the original structural parameters. This slight deviation of TMDI parameters results in a smaller equivalent damping of TMDI and thereby leading to a worse control efficiency.

Figure 11 shows the response of the TMDI mass block. The TMDI has the best control efficiency around the original structural frequency, where the deck response can be well suppressed and the TMDI response is thus also quite small. When the structural frequency departs from the original value at which the TMDI is designed, the equivalent damping of TMDI supplied to the bridge decreases. Thus, the VIV response of the bridge increases. Accordingly, the TMDI response also increases. When the structural frequency further varies from the original value (e.g., $\pm 5\%$), the TMDI becomes detuning to the bridge and gradually loses its efficiency. In this situation, the structural response keeps increasing, but the TMDI response gradually decreases.

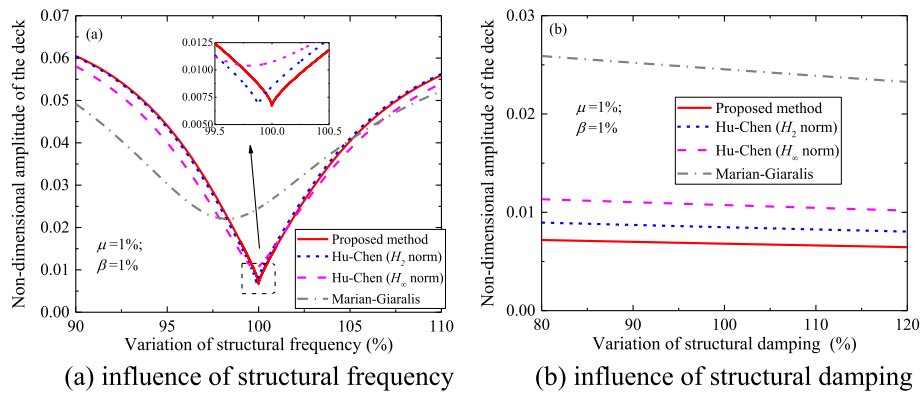


FIGURE 10 Sensitivity of deck response against variations in structural parameters

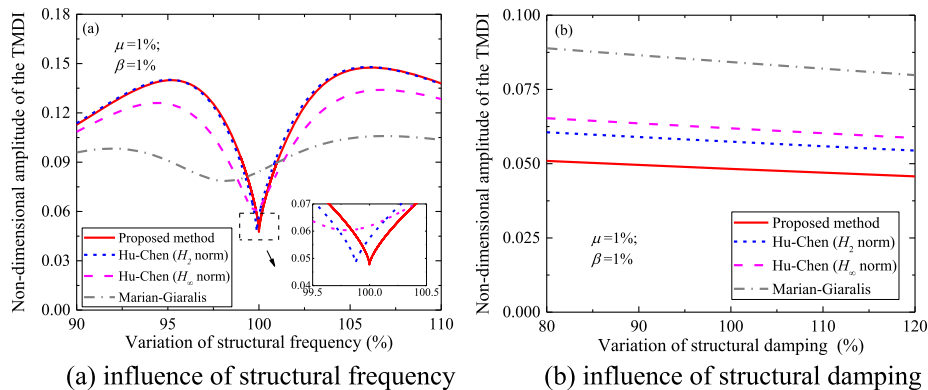


FIGURE 11 Sensitivity of TMDI response against variations in structural parameters

5.2 | Application scope of the design formulas

In the derivation of TMDI design formulas, it is assumed that the influence of TMDI on the natural frequency of the bridge is neglectable ($\gamma = 1$). However, former studies (e.g., Chen et al⁴³) indicated that the inerter can reduce the natural frequency of the main structure at certain circumstances. To validate the reliability of the proposed formulas, the closed-form design formulas are compared with the numerical optimization method. The numerical optimization adopts the similar idea with that in TMD optimization by Dai et al¹⁵: for a given set of TMDI parameters, the value of γ is firstly calculated through solving Equation (13.2); then, the equivalent damping of TMDI ξ_{eq} is calculated by Equation (13.1). If γ has more than one real roots, the root that leads to the smallest equivalent damping of TMDI is considered as the real value. The values of Ω and ξ_t that lead to the maximum value of ξ_{eq} is taken as the optimum design values.

The calculated equivalent damping of TMDI at different mass ratios, inertance ratios, and inerter span lengths is shown in Figure 12. Two mass ratios are considered: $\mu = 0$ corresponds to the TID device, and $\mu = 0.05$ is commonly treated as an upper limit of mass ratio for bridge vibration control. $1 - \Delta\phi = 0$ denotes that the inerter spans zero length. $1 - \Delta\phi = 1$ denotes that the other terminal of inerter is linked to a fixed point of modal shape function (medium inerter span length). And $1 - \Delta\phi = 2$ denotes that the other terminal of inerter is linked to a spanwise position which has an opposite modal shape value with the generalized coordinate (large inerter span length). The solid lines denote the results calculated by the design formulas, and the dashed lines denote the results of numerical optimization.

As can be seen, the equivalent damping ratio of TMDI increases with both the increment of inertance ratio and inerter span length for $\mu = 0$. While for $\mu \neq 0$, there exists a turning point of inerter span length, based on which the influence of inerter can be divided into two categories: when the inerter span length is small ($1 - \Delta\phi < 0.4$), the equivalent damping of TMDI decreases with the increment of inertance ratio β ; beyond that point, the equivalent damping of TMDI increases with the increment of inertance ratio β . The calculated results by design formulas are consistent with the results of numerical optimization at small and medium inertance ratios ($\beta < 0.1$) or inerter span lengths ($1 - \Delta\phi < 0.8$). However, once the inertance ratio β is larger than 0.1 and the inerter span lengths ($1 - \Delta\phi$) is larger than 0.8, the calculated results by design formulas (solid lines) become smaller than those of numerical optimization (dashed lines).

Figure 13 shows the frequency ratio of the bridge with and without TMDI, that is, $\gamma = \omega/\omega_b$. As mentioned above, the proposed design formulas are derived based on the assumption of $\gamma = 1$. In another word, the values of γ for design formulas always equal to 1. Thus, only the results of numerical optimization are shown. Basically, the larger the inerter span length or the inertance ratio, the smaller the values of γ . Within the regions where the results of the two methods agree well in Figure 12, the values of γ are also close to one. This result can further validate the basic assumption behind the proposed formulas and their application scopes.

Normally, the mass ratio of TMD for bridge vibration control is smaller than that for building vibration control (for bridge, μ is normally less than 2%). In this sense, the inertance ratio in TMDI β will not exceed 0.1, since an inertance-to-mass ratio of $\beta/\mu = 5$ can reduce the static deformation of TMDI springs to less than 20% of that in TMD. In addition, long-span bridge is a flexible structure, the difference in modal shape functions is obvious only if the spanwise distance

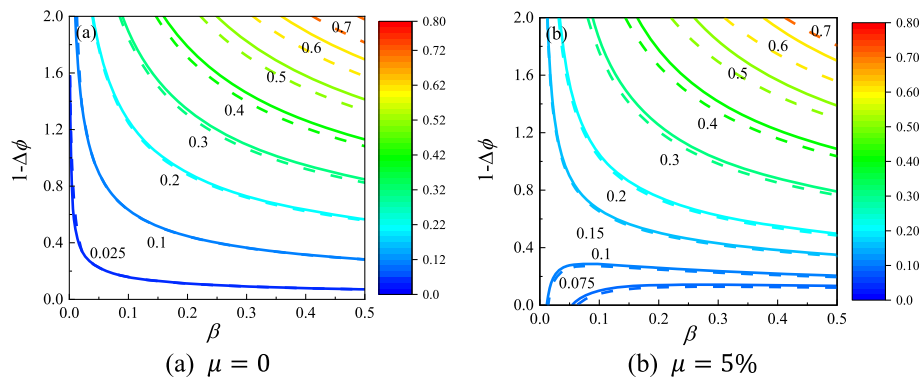


FIGURE 12 Contour lines of TMDI equivalent damping ratios calculated by design formulas and numerical optimization, solid lines: design formulas; dash lines: numerical optimization

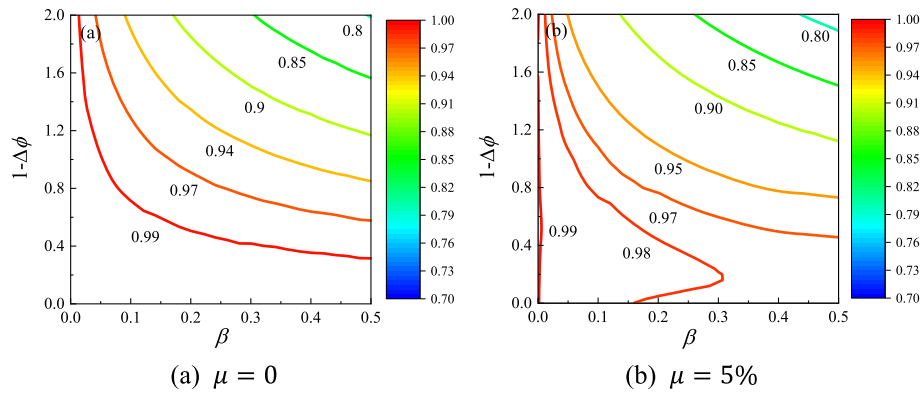


FIGURE 13 Contour lines of frequency ratio of the bridge with and without TMDI

is large enough. In real application, it is hard to span the two terminals of the inerter to that long distance. Thus, the inerter span length value ($1 - \Delta\phi$) can hardly exceed 0.8. Based on these, we can conclude that the real application scope of TMDI is within the range of $0 < \beta < 0.1$ and $0 < 1 - \Delta\phi < 0.8$. In this scope, the design formulas proposed in this study have the same accuracy as numerical optimization and can be used to quickly and conveniently obtain the TMDI parameters.

6 | CONCLUSIONS

Closed-form design formulas of TMDI for suppressing VIV of bridges are proposed in this study. The proposed formulas can consider the influence of structural damping of the bridge and the installation arrangement of TMDI (spanwise location and inerter span length). The accuracy and efficiency of the proposed formulas are validated through experimental data as well as comparison with existing TMDI design formulas. The main findings and results of this study are as follows:

1. The effect of TMDI to the bridge can be treated as directly increasing the mechanical damping of the bridge. When the increased amount of mechanical damping of the bridge equals to the equivalent damping of TMDI, the VIV response of the bridge is consistent with the bridge-TMDI system.
2. Compared to existing design formulas of TMDI, the proposed formulas can obtain a higher equivalent damping for the same predetermined TMDI parameters. In other words, for the same value of required equivalent damping of TMDI, the required mass ratio by using the proposed method can be smaller than those obtained by using existing formulas. The bridge-TMDI system designed by the proposed method has a similar robustness with those designed by formulas of H_2 and H_∞ norm optimizations, but the control efficiency is slightly better.
3. The design formulas of TMDI are independent with VIV force models. However, if the VIV force model inaccurately predicts the required increment of Scruton number by TMDI, significant design errors will occur. To eliminate this adverse influence, it is recommended to obtain the VIV amplitude curve versus Scruton number by wind tunnel experiment or modeling the aeroelastic parameters in VIV models as functions of Scruton number.
4. In real application, the inertance ratio as well as the inerter span length of TMDI are relatively small. In this scope, the proposed design formulas have high accuracy and can be used to quickly and conveniently obtain the TMDI parameters.

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AUTHOR CONTRIBUTION

Kun Xu: Methodology; analysis; writing. **Qian Dai:** Calculation; investigation; review and editing. **Kaiming Bi:** Validation; review and editing. **Genshen Fang:** Validation; review and editing. **Yaojun Ge:** Supervision; wind tunnel experimental resources.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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